Discrete Data Problems

1 Discrete Random Variable with Two Categories: Binomial Model

1.1 Bernoulli trials and the binomial statistic

Many statistical problems involve repeated experiments with only two possible outcomes:

success/failure, yes/no, event/no event.

The fundamental model for such data is the Bernoulli distribution.

Let $X_1, \ldots, X_n$ be independent Bernoulli(p) trials, where

$$\mathbb{P}(X_i = 1) = p, \qquad \mathbb{P}(X_i = 0) = 1 - p.$$

Thus, X_i is an indicator of success on trial i.

Mean and variance. A basic calculation gives

$$\mathbb{E}(X_i) = p, \qquad \mathrm{Var}(X_i) = p(1-p).$$

The variance is largest at $p = 1/2$, reflecting maximum uncertainty.

Binomial count of successes. The total number of successes is

$$B = \sum_{i=1}^{n} X_i.$$

Since the trials are independent and identically distributed,

$$B \sim \mathrm{Binomial}(n, p),$$

with probability mass function

$$\mathbb{P}(B = b) = \binom{n}{b} p^b (1-p)^{n-b}, \qquad b = 0, 1, \ldots, n.$$

Interpretation. The binomial model is the discrete analogue of many "counting" procedures in nonparametrics, including the sign test, where we count positive versus negative signs.

Likelihood viewpoint. The binomial likelihood for p is

$$L(p \mid b) = \binom{n}{b} p^b (1-p)^{n-b},$$

and the maximum likelihood estimator is

$$\hat{p} = \frac{B}{n}.$$

1.2 Exact binomial test

Suppose we want to test whether the success probability equals some benchmark value p_0:

$$H_0 : p = p_0 \qquad \text{vs} \qquad H_A : p \neq p_0,$$

with one-sided alternatives defined similarly.

Under H_0, the distribution of B is exactly known:

$$B \sim \text{Binomial}(n, p_0).$$

Exact p-value. If we observe $B = b_{\text{obs}}$, then:

- For $H_A : p > p_0$,
$$p\text{-value} = \mathbb{P}(B \geq b_{\text{obs}}).$$
- For $H_A : p < p_0$,
$$p\text{-value} = \mathbb{P}(B \leq b_{\text{obs}}).$$
- For $H_A : p \neq p_0$, the p-value is based on both tails, typically doubling the smaller tail probability.

Rejection regions (exact). Let $b_{\alpha;n,p_0}$ denote an upper α critical value:

- For $H_A : p > p_0$, reject if $B \geq b_{\alpha;n,p_0}$.
- For $H_A : p < p_0$, reject if $B \leq c_{\alpha;n,p_0}$.
- For $H_A : p \neq p_0$, reject in both tails:
$$B \geq b_{\alpha_1;n,p_0} \quad \text{or} \quad B \leq c_{\alpha_2;n,p_0}, \qquad \alpha_1 + \alpha_2 = \alpha.$$

Remark 1.1 (Discrete nature of exact tests). Because B takes only integer values, it is not always possible to construct a test with size exactly equal to a chosen α. Instead, the achieved significance level may be slightly smaller than α. This is why exact tests can be conservative.

Remark 1.2 (Asymmetric null distribution). When $p_0 \neq 1/2$, the binomial null distribution is skewed. In this case, equal splitting $\alpha_1 = \alpha_2 = \alpha/2$ does not generally maximize power.

For two-sided testing in exponential families, the uniformly most powerful unbiased (UMPU) test requires the additional unbiasedness condition

$$\mathbb{E}_{p_0}\left[(B - np_0)\mathbf{1}_{\text{reject}}\right] = 0,$$

which typically leads to unequal tail probabilities.

Thus, when the null distribution is asymmetric, optimal allocation of α_1 and α_2 depends on whether one desires

- symmetry (balanced sensitivity),
- unbiasedness (UMPU optimality), or
- increased power in a scientifically preferred direction.

Remark 1.3 (Choosing α_1 by a finite search). Because B is discrete, the mappings $\alpha_1 \mapsto b_{\alpha_1}$ and $\alpha_2 \mapsto c_{\alpha_2}$ are step functions. Hence optimizing over $\alpha_1 \in [0, \alpha]$ is equivalent to searching over a finite set of attainable critical values.

If one targets power at a specific alternative p_1, a natural criterion is

$$\pi(p_1; \alpha_1) = \mathbb{P}_{p_1}(B \geq b_{\alpha_1}) + \mathbb{P}_{p_1}(B \leq c_{\alpha-\alpha_1}),$$

subject to the size constraint

$$\mathbb{P}_{p_0}(B \geq b_{\alpha_1}) + \mathbb{P}_{p_0}(B \leq c_{\alpha-\alpha_1}) \leq \alpha.$$

For $p_1 > p_0$, the maximizer typically allocates more of α to the upper tail. If instead one desires a two-sided test with formal optimality, one can search over tail cutoffs to satisfy the UMPU (unbiasedness) condition in addition to size.

1.3 Large-sample approximation for the binomial test

When n is large, the binomial distribution is well approximated by a normal distribution.

Mean and variance under H_0.

$$\mathbb{E}(B) = np_0, \qquad \mathrm{Var}(B) = np_0(1 - p_0).$$

Define the standardized statistic

$$Z = \frac{B - np_0}{\sqrt{np_0(1 - p_0)}}.$$

Central Limit Theorem. Under H_0, as $n \to \infty$,

$$Z \overset{approx}{\sim} N(0, 1).$$

Equivalently,

$$Z^2 \approx \chi^2_1,$$

which connects the binomial test to Pearson chi-square methods.

Rejection regions (normal approximation).

- $H_A : p > p_0$: reject if $Z \geq z_\alpha$.
- $H_A : p < p_0$: reject if $Z \leq -z_\alpha$.
- $H_A : p \neq p_0$: reject if $|Z| \geq z_{\alpha/2}$.

Continuity correction. Since B is discrete but the normal distribution is continuous, a continuity correction sometimes improves accuracy:

$$Z = \frac{B + 0.5 - np_0}{\sqrt{np_0(1 - p_0)}}.$$

Remark 1.4 (When is the approximation valid?). The normal approximation works best when both expected counts are not too small:

$$np_0 \gtrsim 5, \qquad n(1 - p_0) \gtrsim 5.$$

If p_0 is very close to 0 or 1, exact methods are preferred.

1.4 Simulation illustrations

Simulation helps visualize the transition from discrete binomial to normal shape.

Small sample ($n = 10$).

```
n = 10;  p0 = 1/2;  nsim = 1000
B = rbinom(nsim, size = n, prob = p0)

hist(B, breaks = 0:10,
     main="Binomial(n=10, p=0.5)",
     xlab="Number of successes")

qqnorm(B); qqline(B)
```

Larger sample ($n = 100$).

```
n = 100; p0 = 1/2;  nsim = 10000
B = rbinom(nsim, size = n, prob = p0)

hist(B, breaks = 30,
     main="Binomial(n=100, p=0.5)",
     xlab="Number of successes")

qqnorm(B); qqline(B)
```

As n increases, the histogram becomes smoother and more bell-shaped, illustrating the CLT in action.

2 Example: Sensory Difference Tests (Triangle Test)

2.1 Design

Triangle tests are widely used in food science and sensory evaluation.

- Each of n panelists receives three samples in random order.
- Two samples are identical, one is different.
- The panelist must identify the odd sample.

Under pure guessing, the success probability is

$$p_0 = \frac{1}{3}.$$

We test

$$H_0 : p = \frac{1}{3} \qquad \text{vs} \qquad H_A : p > \frac{1}{3}.$$

2.2 Large-sample test statistic

Let B be the number of correct identifications. Then under H_0,

$$B \sim \text{Binomial}\Big(n, \tfrac{1}{3}\Big).$$

The standardized statistic is

$$Z = \frac{B - n(1/3)}{\sqrt{n(1/3)(2/3)}}.$$

Example. With $n = 50$ and observed $B = 25$,

$$Z_{\text{obs}} = \frac{25 - 50(1/3)}{\sqrt{50(1/3)(2/3)}} = 2.5.$$

At $\alpha = 0.05$ one-sided, reject if $Z \geq 1.645$. Thus we reject H_0.

2.3 p-value: exact vs approximate

Approximate p-value:

$$p_{\text{approx}} = 1 - \Phi(2.5) \approx 0.0062.$$

Exact p-value:

$$p_{\text{exact}} = \mathbb{P}(B \geq 25), \quad B \sim \text{Binomial}(50, 1/3) \approx 0.0108.$$

Remark 2.1. The discrepancy occurs because the normal approximation is imperfect for moderate sample sizes and skewed probabilities like $p_0 = 1/3$. Exact binomial inference remains the gold standard when feasible.

3 Power Calculation (Normal Approximation)

Power is the probability of rejecting H_0 when the alternative H_A is true.

Suppose we test

$$H_0 : p = p_0 \qquad \text{vs} \qquad H_A : p > p_0,$$

where $B \sim \text{Binomial}(n, p)$.

Using the normal approximation under H_0, the standardized statistic is

$$Z = \frac{B - np_0}{\sqrt{np_0(1 - p_0)}}.$$

At significance level $\alpha = 0.05$ (one-sided),

$$z_{0.05} = 1.645.$$

Thus we reject H_0 when

$$Z \geq 1.645.$$

This is equivalent to rejecting when

$$B \geq np_0 + 1.645\sqrt{np_0(1-p_0)}.$$

Power is computed under the true value p (for example $p = 0.6$):

$$\text{Power}(p) = \mathbb{P}_p(\text{Reject } H_0) = \mathbb{P}_p\left(B \geq np_0 + 1.645\sqrt{np_0(1-p_0)}\right).$$

Under the alternative,

$$B \sim \text{Binomial}(n, p),$$

and we again approximate using a normal distribution:

$$B \approx N\big(np, np(1-p)\big).$$

We standardize using the *true* mean and variance:

$$\mathbb{P}_p\left(\frac{B - np}{\sqrt{np(1-p)}} \geq \frac{np_0 + 1.645\sqrt{np_0(1-p_0)} - np}{\sqrt{np(1-p)}}\right).$$

Since

$$\frac{B - np}{\sqrt{np(1-p)}} \approx N(0, 1),$$

the power is approximately

$$\text{Power}(p) \approx 1 - \Phi\left(\frac{np_0 + 1.645\sqrt{np_0(1-p_0)} - np}{\sqrt{np(1-p)}}\right).$$

This expression is the approximate *power function* of the test.

Example

If $p_0 = 0.5$, $p = 0.6$, and $n = 100$,

$$\text{Cutoff} = 50 + 1.645\sqrt{25} = 50 + 8.225 = 58.225.$$

Under $p = 0.6$,

$$\mu = 60, \qquad \sigma = \sqrt{100(0.6)(0.4)} = \sqrt{24} \approx 4.90.$$

Thus

$$\text{Power} \approx \mathbb{P}\left(Z \geq \frac{58.225 - 60}{4.90}\right) = \mathbb{P}(Z \geq -0.36) = \Phi(0.36) \approx 0.64.$$

So the test has roughly 64% power at $p = 0.6$ with $n = 100$.

4 Estimator and Confidence Interval for a Success Probability

4.1 Point estimator

The natural estimator is

$$\hat{p} = \frac{B}{n}.$$

4.2 Standard error

$$\mathrm{se}(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}, \qquad \widehat{\mathrm{se}}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

4.3 Wald confidence interval

$$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

Remark 4.1 (Limitations). The Wald CI performs poorly when n is small or $\hat{p}$ is near 0 or 1. Exact or Wilson intervals are often preferred.

5 Discrete Random Variable with More Than Two Categories

5.1 Pearson Chi-Squared Goodness-of-Fit Test

Let X take k categories with probabilities $p_1, \dots, p_k$.

Observed counts:

$$X_1, \dots, X_k, \qquad \sum X_i = n.$$

Test

$$H_0 : p_i = p_i^0.$$

Pearson statistic:

$$\chi^2 = \sum_{i=1}^{k} \frac{(X_i - np_i^0)^2}{np_i^0}.$$

Under H_0,

$$\chi^2 \xrightarrow{d} \chi^2_{k-1}.$$

Why $k-1$ degrees of freedom? Because probabilities satisfy $\sum p_i = 1$, one category is determined by the others.

Pearson χ^2 Goodness-of-Fit from the Multinomial CLT Let $\boldsymbol{X} = (X_1, \dots, X_k)^\top \sim$ Multinomial(n, p^0) under

$$H_0 : \quad p = p^0 = (p_1^0, \dots, p_k^0)^\top, \qquad p_i^0 > 0, \ \sum_{i=1}^{k} p_i^0 = 1.$$

Define empirical proportions $\hat{\boldsymbol{p}} = \boldsymbol{X}/n$.

Mean and covariance under H_0. For the multinomial,

$$\mathbb{E}(\boldsymbol{X}) = n\boldsymbol{p}^0, \qquad \mathrm{Cov}(\boldsymbol{X}) = n\boldsymbol{\Sigma}_0,$$

where

$$\boldsymbol{\Sigma}_0 = \mathrm{diag}(\boldsymbol{p}^0) - \boldsymbol{p}^0(p^0)^\top.$$

Equivalently,

$$\sqrt{n}\,(\hat{\boldsymbol{p}} - \boldsymbol{p}^0) \text{ has covariance } \boldsymbol{\Sigma}_0.$$

Let $\boldsymbol{e}_m$ be a unit vector with 1 at the m-th position and zero otherwise. Write $\boldsymbol{X} = \sum_{\ell=1}^{n} \boldsymbol{Y}_\ell$ where $\boldsymbol{Y}_\ell \in \{\boldsymbol{e}_1, \ldots, \boldsymbol{e}_k\}$ for ℓ-th sample, with $\mathbb{P}(\boldsymbol{Y}_\ell = \boldsymbol{e}_i) = p_i^0$. Then $\mathbb{E}(\boldsymbol{Y}_\ell) = p^0$ and $\mathrm{Cov}(\boldsymbol{Y}_\ell) = \boldsymbol{\Sigma}_0$, and by the multivariate CLT,

$$\sqrt{n}\,(\hat{\boldsymbol{p}} - \boldsymbol{p}^0) = \frac{1}{\sqrt{n}} \sum_{\ell=1}^{n} (\boldsymbol{Y}_\ell - \boldsymbol{p}^0) \ \xrightarrow{d}\ N_k(0, \boldsymbol{\Sigma}_0).$$

Note Σ_0 is singular with $\mathrm{rank}(\Sigma_0) = k - 1$ because $\sum_i \hat{p}_i = 1$.

Pearson statistic as a quadratic form. Pearson's goodness-of-fit statistic is

$$\chi^2 = \sum_{i=1}^{k} \frac{(X_i - np_i^0)^2}{np_i^0} = \big(\boldsymbol{X} - n\boldsymbol{p}^0\big)^\top \Big(n\,\mathrm{diag}(\boldsymbol{p}^0)\Big)^{-1} \big(\boldsymbol{X} - n\boldsymbol{p}^0\big).$$

In terms of $\hat{\boldsymbol{p}}$,

$$\chi^2 = n(\hat{\boldsymbol{p}} - \boldsymbol{p}^0)^\top \mathrm{diag}(\boldsymbol{p}^0)^{-1} (\hat{\boldsymbol{p}} - \boldsymbol{p}^0).$$

Define

$$\boldsymbol{Z}_n = \sqrt{n}\,\mathrm{diag}(\boldsymbol{p}^0)^{-1/2} (\hat{\boldsymbol{p}} - \boldsymbol{p}^0).$$

Then

$$\chi^2 = \|\boldsymbol{Z}_n\|_2^2 = \boldsymbol{Z}_n^\top \boldsymbol{Z}_n.$$

Limit distribution is χ^2_{k-1}. From the CLT,

$$\boldsymbol{Z}_n \ \xrightarrow{d}\ N_k\Big(0,\ \underbrace{\mathrm{diag}(\boldsymbol{p}^0)^{-1/2} \boldsymbol{\Sigma}_0\, \mathrm{diag}(\boldsymbol{p}^0)^{-1/2}}_{=\boldsymbol{A}}\Big).$$

Compute $\boldsymbol{A}$:

$$\boldsymbol{A} = \mathrm{diag}(\boldsymbol{p}^0)^{-1/2} \Big(\mathrm{diag}(\boldsymbol{p}^0) - \boldsymbol{p}^0 (\boldsymbol{p}^0)^\top\Big) \mathrm{diag}(\boldsymbol{p}^0)^{-1/2} = \boldsymbol{I}_k - \boldsymbol{u}\boldsymbol{u}^\top,$$

where

$$\boldsymbol{u} = \big(\sqrt{p_1^0}, \ldots, \sqrt{p_k^0}\big)^\top, \qquad \|u\|_2^2 = \sum_{i=1}^{k} p_i^0 = 1.$$

Thus $\boldsymbol{A}$ is an idempotent projection matrix: $\boldsymbol{A}^2 = \boldsymbol{A}$, with

$$\mathrm{rank}(\boldsymbol{A}) = k - 1,$$

since it projects onto the $(k-1)$-dimensional subspace orthogonal to u.

Hence,

$$\chi^2 = \boldsymbol{Z}_n^\top \boldsymbol{Z}_n \ \xrightarrow{d} \chi^2_{k-1}.$$

Thus, Under H_0 and $p_i^0 > 0$,

$$\sum_{i=1}^{k} \frac{(X_i - np_i^0)^2}{np_i^0} \ \xrightarrow{d}\ \chi^2_{k-1}.$$

Rule of thumb (validity). The χ^2 approximation is typically accurate when expected counts np_i^0 are not too small (often $np_i^0 \gtrsim 5$ for all i).

6 Example: Mendel's Pea Plant Experiments (3:1 Ratio)

Mendel's pea plant experiments provide one of the most classical and historically important examples of a *goodness-of-fit* (GOF) test.

Mendel studied traits such as flower color, seed shape, and pod color. For each trait, the offspring were classified into two categories:

- **Dominant phenotype** (e.g., purple flowers),
- **Recessive phenotype** (e.g., white flowers).

Mendelian prediction: the 3:1 ratio

Under simple Mendelian inheritance (one gene with two alleles), the expected probabilities are

$$p_{\text{dom}} = \frac{3}{4} = 0.75, \qquad p_{\text{rec}} = \frac{1}{4} = 0.25.$$

Suppose for a given trait we observe counts

$$O_{\text{dom}}, \qquad O_{\text{rec}}, \qquad \text{with total } n = O_{\text{dom}} + O_{\text{rec}}.$$

Under the null hypothesis H_0 (Mendel's genetic model), the *expected* counts are

$$E_{\text{dom}} = 0.75n, \qquad E_{\text{rec}} = 0.25n.$$

Chi-square goodness-of-fit test

To measure how close the observed counts are to the expected Mendelian ratio, we compute the chi-square GOF statistic:

$$\chi^2 = \frac{(O_{\text{dom}} - E_{\text{dom}})^2}{E_{\text{dom}}} + \frac{(O_{\text{rec}} - E_{\text{rec}})^2}{E_{\text{rec}}}.$$

Because there are two categories and no parameters are estimated from the data, this statistic has approximately a chi-square distribution with

$$\text{df} = 2 - 1 = 1.$$

Thus, under H_0,

$$\chi^2 \sim \chi^2_1.$$

Combining evidence across multiple traits

Mendel reported results across several traits (for example, 7 traits). For each trait j, one can compute a separate statistic:

$$\chi^2_{1,j}.$$

If the traits are approximately independent, then the sum satisfies

$$\sum_{j=1}^{7} \chi^2_{1,j} \sim \chi^2_7.$$

This provides an overall test of whether Mendel's results agree with the theoretical Mendelian ratios across all traits.

When is the chi-square approximation valid?

The chi-square approximation works well when expected counts are not too small. A common rule of thumb is:

$$E_{\text{dom}}, E_{\text{rec}} \geq 5.$$

If expected counts are very small, exact binomial tests or simulation-based methods may be preferred.

Interpretation

- A small χ^2 value indicates close agreement with Mendel's prediction.
- A large χ^2 value suggests deviation from the 3:1 ratio, possibly due to chance, experimental issues, or model failure.

This example highlights how chi-square GOF tests provide a quantitative way to compare observed categorical data with theoretical probability models.

7 Testing Two Success Probabilities: 2×2 Tables

A very common problem in biostatistics and clinical research is comparing two success probabilities:

- group 1 has success probability p_1,
- group 2 has success probability p_2.

Examples include:

- treatment vs placebo response rates,
- disease prevalence in exposed vs unexposed groups,
- survival vs death proportions in two populations.

The 2×2 table setup

Data are summarized in a 2×2 contingency table:

	Success	Failure	Total
Group 1	X_1	$n_1 - X_1$	n_1
Group 2	X_2	$n_2 - X_2$	n_2
Total	$X_1 + X_2$	$n - (X_1 + X_2)$	n

The sample proportions are

$$\hat{p}_1 = \frac{X_1}{n_1}, \qquad \hat{p}_2 = \frac{X_2}{n_2}.$$

The main hypothesis of interest is

$$H_0 : p_1 = p_2 \qquad \text{vs} \qquad H_a : p_1 \neq p_2.$$

7.1 Fisher's exact test

When sample sizes are small, the usual chi-square approximation may not be accurate, especially if some expected cell counts are less than 5.

Under H_0, conditional on the margins (n_1, n_2, S), the $S = X_1 + X_2$ successes are randomly distributed among the n individuals.

Key idea (conditioning on margins). Fix the row totals (n_1, n_2) and the total number of successes S. Then

$$X_1 = \text{number of successes that fall in group 1.}$$

This is sampling without replacement:

- population size: $N = n$,
- number of successes in population: $K = S$,
- sample size (group 1 size): n_1,
- random variable: X_1.

Therefore,

$$X_1 \mid (S, n_1, n) \sim \text{Hypergeometric}(N = n, K = S, n_1).$$

Derivation To compute $\mathbb{P}(X_1 = x)$:

- choose x successes from the S total successes,
- choose $n_1 - x$ failures from the $F = n - S$ failures,
- divide by the total number of ways to choose n_1 individuals from n.

Hence,

$$\mathbb{P}(X_1 = x \mid S, n_1, n) = \frac{\binom{S}{x}\binom{n-S}{n_1-x}}{\binom{n}{n_1}}.$$

Substituting $S = X_1 + X_2$ gives

$$\mathbb{P}(X_1 = x) = \frac{\binom{X_1+X_2}{x}\binom{n-(X_1+X_2)}{n_1-x}}{\binom{n}{n_1}}.$$

Support. The allowable values of x satisfy

$$\max\big(0, n_1 - (n - S)\big) \le x \le \min(n_1, S).$$

In this setting, Fisher's exact test provides an *exact* test of association.

Exact p-value

The p-value is computed by summing probabilities of all tables that are as extreme or more extreme than the observed one.

- Fisher's test is valid for any sample size.
- It is especially recommended when expected counts are small.

When is Fisher's test used?

- Small clinical trials,
- Rare outcomes,
- Sparse contingency tables.

7.2 Large-sample z-test for $p_1 - p_2$

When sample sizes are sufficiently large, we can use a normal approximation to compare two proportions.

The difference in sample proportions is

$$\hat{p}_1 - \hat{p}_2.$$

Pooled estimator under the null

Under the null hypothesis $H_0 : p_1 = p_2 = p$, we estimate the common probability by pooling both groups:

$$\hat{p}_d = \frac{X_1 + X_2}{n_1 + n_2}.$$

Test statistic

The large-sample z-statistic is

$$Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}_d(1 - \hat{p}_d)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}.$$

Under H_0, we have approximately

$$Z \sim N(0, 1).$$

Thus, a two-sided p-value is

$$p\text{-value} = 2P(|Z| > |z_{\text{obs}}|).$$

Connection to chi-square

Squaring the z-statistic gives

$$Z^2 \approx \chi^2_1,$$

so this test is equivalent to the Pearson chi-square test in the 2×2 case.

When is the z-test appropriate?

A common rule of thumb is that all expected counts should satisfy

$$E_{ij} \geq 5.$$

8 2×2 Pearson Chi-Squared Tests

Another standard approach to comparing two proportions is the Pearson chi-square test for independence in a 2×2 contingency table.

Expected counts under independence

Let O_{ij} denote the observed cell counts.

Under the null hypothesis of independence, the expected counts are

$$E_{ij} = \frac{n_{i\cdot} n_{\cdot j}}{n},$$

[why? Ideally under independence, we should have $p_{ij} = p_i p_j$. Thus, $\frac{E_{ij}}{n} = \frac{n_{\cdot i}}{n} \frac{n_{\cdot j}}{n}$ which simplifies to above result.]

where

- $n_{i\cdot}$ are the row totals,
- $n_{\cdot j}$ are the column totals,
- n is the grand total.

Pearson chi-square statistic

The Pearson statistic is

$$\chi^2 = \sum_{i=1}^{2} \sum_{j=1}^{2} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}.$$

Under H_0 and for large samples,

$$\chi^2 \approx \chi^2_1.$$

Interpretation

- Small χ^2 means observed counts match independence.
- Large χ^2 provides evidence of association between group and outcome.

Continuity correction (Yates). Because the chi-square distribution is continuous while the data are discrete, the Pearson approximation can slightly overstate significance in small samples.

Yates' continuity correction adjusts for this by subtracting 0.5:

$$\chi^2_{\text{Yates}} = \sum_{i,j} \frac{(|O_{ij} - E_{ij}| - 0.5)^2}{E_{ij}}.$$

Effect of the correction

- Makes the test more conservative (larger p-values),
- Mainly used when counts are small,
- Often replaced today by Fisher's exact test in sparse tables.

9 More Than One Discrete Random Variable: Contingency Tables

So far, we have focused on goodness-of-fit tests for a *single* categorical random variable.

In many applications, however, we observe *two* discrete random variables simultaneously, and we want to study whether they are related.

This leads to the framework of *contingency tables*.

Two-way tables

A contingency table records counts for two categorical variables:

- X takes values in categories $i = 1, \dots, r$,
- Y takes values in categories $j = 1, \dots, c$.

The observed count in cell (i, j) is

$$O_{ij} = \#\{(X, Y) \in (i, j)\}.$$

All counts together form an $r \times c$ table:

	Y_1	Y_2	$\cdots$	Row total
X_1	O_{11}	O_{12}	$\cdots$	$n_{1\cdot}$
X_2	O_{21}	O_{22}	$\cdots$	$n_{2\cdot}$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
X_r	O_{r1}	O_{r2}	$\cdots$	$n_{r\cdot}$
Column total	$n_{\cdot 1}$	$n_{\cdot 2}$	$\cdots$	n

Marginal totals

The row totals are

$$n_{i\cdot} = \sum_{j=1}^{c} O_{ij},$$

and the column totals are

$$n_{\cdot j} = \sum_{i=1}^{r} O_{ij}.$$

The grand total sample size is

$$n = \sum_{i=1}^{r} \sum_{j=1}^{c} O_{ij}.$$

Testing independence

A central question in contingency table analysis is:

Are X and Y independent?

The null hypothesis is

$$H_0 : X \text{ and } Y \text{ are independent.}$$

Under independence, the joint probability factors:

$$P(X = i, Y = j) = P(X = i)\, P(Y = j).$$

This implies expected counts of the form

$$E_{ij} = \frac{n_{i\cdot}\, n_{\cdot j}}{n}.$$

Chi-square test for contingency tables

To compare observed and expected counts, we compute the chi-square statistic:

$$\chi^2 = \sum_{i=1}^{r} \sum_{j=1}^{c} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}.$$

Under H_0, this statistic is approximately chi-square distributed with

$$\mathrm{df} = (r-1)(c-1).$$

Interpretation

- If χ^2 is small, the data are consistent with independence.
- If χ^2 is large, there is evidence that X and Y are associated.

Contingency table methods generalize goodness-of-fit ideas to the study of relationships between multiple categorical variables.

10 Measures and Visualizations of Association

The chi-square test tells us whether there is evidence against independence, but it does not explain *where* the association comes from.

In practice, after rejecting H_0, we want to answer:

Which cells contribute most to the departure from independence?

This motivates residual-based diagnostics and visualization tools.

10.1 Residual-based interpretation

Recall that the Pearson chi-square statistic is

$$\chi^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}.$$

This shows that the overall test statistic is built from cell-by-cell contributions.

Pearson residuals

A convenient standardized measure of deviation in each cell is the *Pearson residual*:

$$r_{ij} = \frac{O_{ij} - E_{ij}}{\sqrt{E_{ij}}}.$$

These residuals compare the observed count to the expected count under independence, scaled by the standard deviation $\sqrt{E_{ij}}$.

Interpretation

- If $r_{ij} \approx 0$, then the cell count is close to what independence predicts.
- If $r_{ij} > 0$, then
$$O_{ij} > E_{ij},$$
meaning there are *more observations than expected* in that category combination.
- If $r_{ij} < 0$, then
$$O_{ij} < E_{ij},$$
meaning there are *fewer observations than expected*.

Which cells drive the association?

Cells with large absolute residuals,

$$|r_{ij}| \gg 0,$$

are the main contributors to the chi-square statistic and therefore drive the rejection of independence.

As a rough guideline:

- $|r_{ij}| > 2$ suggests a notable departure,
- $|r_{ij}| > 3$ suggests a strong departure.

Thus, residuals provide a diagnostic tool beyond the single p-value.

10.2 Association plots

While residuals can be examined numerically, it is often more informative to visualize them.

Association plots provide a graphical representation of the deviations between observed and expected cell counts.

Main idea

Under the null hypothesis of independence:

$$O_{ij} \approx E_{ij}, \qquad r_{ij} \approx 0,$$

so the plot should show bars close to zero.

When independence fails, some cells will show large positive or negative residuals, highlighting the source of association.

R code example

In R, the function `assocplot()` produces an association plot with other parts applied to this example (Example HWC 10.2, page 503):

We consider the following 2×2 contingency table:

	Favor Gun Reg	Oppose Gun Reg
Favor Death Penalty	784	236
Oppose Death Penalty	311	66

Chi-square test of independence We test

$$H_0: \text{ Death penalty and gun registration are independent.}$$

The Pearson chi-square statistic is

$$\chi^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}},$$

where

$$E_{ij} = \frac{(\text{row total}) \times (\text{column total})}{n}.$$

In R:

```
Table10.4 = data.frame(favor = c(784,311),
                       oppose = c(236,66))
chisq.test(Table10.4, correct = FALSE)
```

The option `correct = FALSE` removes Yates' continuity correction.

Expected counts The expected counts under H_0 are obtained by

```
chisq.test(Table10.4, correct = FALSE)$expected
```

These are computed using

$$E_{ij} = \frac{(\text{row total})(\text{column total})}{n}.$$

Association plot To visualize the deviations from independence:

```
x = margin.table(as.matrix(Table10.4), c(1,2))
assocplot(x,
          xlab = "death penalty",
          ylab = "gun registration",
          main = "Gun registration and death penalty")
```

Interpretation of the association plot Each rectangle:

- has an area proportional to the observed count,
- has height proportional to the Pearson residual

$$r_{ij} = \frac{O_{ij} - E_{ij}}{\sqrt{E_{ij}}}.$$

- appears above the baseline if $O_{ij} > E_{ij}$,
- appears below the baseline if $O_{ij} < E_{ij}$.

Thus, the association plot is a graphical decomposition of the chi-square statistic, since each cell contributes

$$\frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

to the overall test statistic.

Here, `x` is a contingency table object.

How to read an association plot

- Bars above zero indicate **excess counts**:

$$O_{ij} > E_{ij}.$$

- Bars below zero indicate **deficits**:

$$O_{ij} < E_{ij}.$$

- The height of the bar is proportional to the magnitude of the residual.
- Large residual patterns reveal which category combinations are most strongly associated.

Connection to hypothesis testing

- Under the null hypothesis of independence, residuals should fluctuate randomly around zero.
- In real data, large systematic residuals indicate departures from independence, leading to rejection of H_0.
- Association plots help interpret the chi-square test by showing *where* the dependence occurs.

10.3 Mosaic plots as an alternative visualization

Another popular visualization for contingency tables is the *mosaic plot*.

- The area of each tile represents the observed frequency.
- Deviations from independence can be highlighted by shading.
- Mosaic plots are especially useful for larger tables.

Thus, graphical tools such as association plots and mosaic plots provide important insight beyond the overall chi-square test result.